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PMC7252948 | 光氧化还原催化 | 采集状态 imported_unchecked

标题	Umpolung Difunctionalization of Carbonyls via Visible-Light Photoredox Catalytic Radical-Carbanion Relay
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出处	Journal of the American Chemical Society 2020
DOI	10.1021/jacs.0c00629
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC7252948/
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摘要（原文节选）

The combination of photoredox catalysis with the Wolff – Kishner (WK) reaction allows the difunctionalization of carbonyl groups by a radical-carbanion relay sequence (photo-Wolff – Kishner reaction). Photoredox initiated radical addition to N-sulfonylhydrazones yields α -functionalized carbanions following the WK-type mechanism. With sulfur-centered radicals, the carbanions are further functionalized by reaction with electrophiles including CO2 and aldehydes, whereas CF3 radical addition furnishes a wide range of gem-d ⋯

数据表 · Table 1（8 行，含数值行 8 行）

Screening of Reaction Conditions for Thiocarboxylation of N-Tosylhydrazonea

entry	change from standard conditions	yieldb
1	none	81%
2	one-pot process	80%c
3	MeCN instead of DMSO	n.d.
4	THF instead of DMSO	n.d.
5	4CzIPN instead of Ir – F	n.d.
6	without Cs2CO3	n.d.
7	without PC	n.d.
8	in the dark	n.d.

脚注（逐字）：aReaction conditions: compound 1a (0.2 mmol), 2a (0.3 mmol), Cs2CO3 (0.6 mmol), [Ir(dFCF3ppy)2dtbbpy]PF6 (1 mol %), and 3 atm of CO2 in 2 mL of solvent, irradiation with blue LED (455 nm) at 25 ° C for 24 h. n.d. = not detected.bYields were determined by 1H NMR analysis of the crude reaction mixture

正文对应句（逐字）：Reaction conditions: unless otherwise noted, all reactions were carried out with 1 (0.2 mmol), 2a (0.3 mmol), Cs2CO3 (0.6 mmol), [Ir(dFCF3ppy)2dtbbpy]PF6 (1 mol %), and 3 atm of CO2 in 2 mL of DMSO, irradiation with blue LED (455 nm) at 25 ° C for 24 h, and isolated yields were shown.

数据表 · Table 5（6 行，含数值行 6 行）

Screening of Reaction Conditions for the 1,1-Difluoroolefination of N-Tosylhydrazonea

entry	change from standard conditions	yieldb
1	none	70%
2	PC (2 mol %)	77% (73%)c
3	one-pot, PC (2 mol %)	75%d
4	without Cs2CO3	n.d.
5	without PC	n.d.
6	in the dark	n.d.

脚注（逐字）：aReaction conditions: compound 7a (0.2 mmol), 8 (0.3 mmol), Cs2CO3 (0.3 mmol), [Ir(dFCF3ppy)2dtbbpy]PF6 (1 mol %) in 1 mL of solvent, irradiation with blue LED (455 nm) at 25 ° C for 24 h. n.d. = not detected.bYields were determined by 19F NMR analysis of the crude reaction mixture using 4,4 ′ -difluor

正文对应句（逐字）：Reaction conditions: unless otherwise noted, all reactions were carried out with 1 (0.2 mmol), 2a (0.3 mmol), Cs2CO3 (0.6 mmol), [Ir(dFCF3ppy)2dtbbpy]PF6 (1 mol %), and 3 atm of CO2 in 2 mL of DMSO, irradiation with blue LED (455 nm) at 25 ° C for 24 h, and isolated yields were shown.

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